

EXERCISES AND EXACT SOLUTIONS

Core Agricultural Indicators With Denominators

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C09
Notebook	V7_A_N04
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Distinguish mean and ratio yield.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Add confidence intervals.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Explain why indicators cannot set budgets.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Mean yield averages holding yields; ratio yield divides estimated total production by total area and answers a different question.

Solution 2

Use the survey design, clustering, stratification, and weights.

Solution 3

Budgets also require objectives, costs, equity, constraints, alternatives, authority, and uncertainty.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated